Implementing and training a diffusion-style drafter

From scratch: learnings

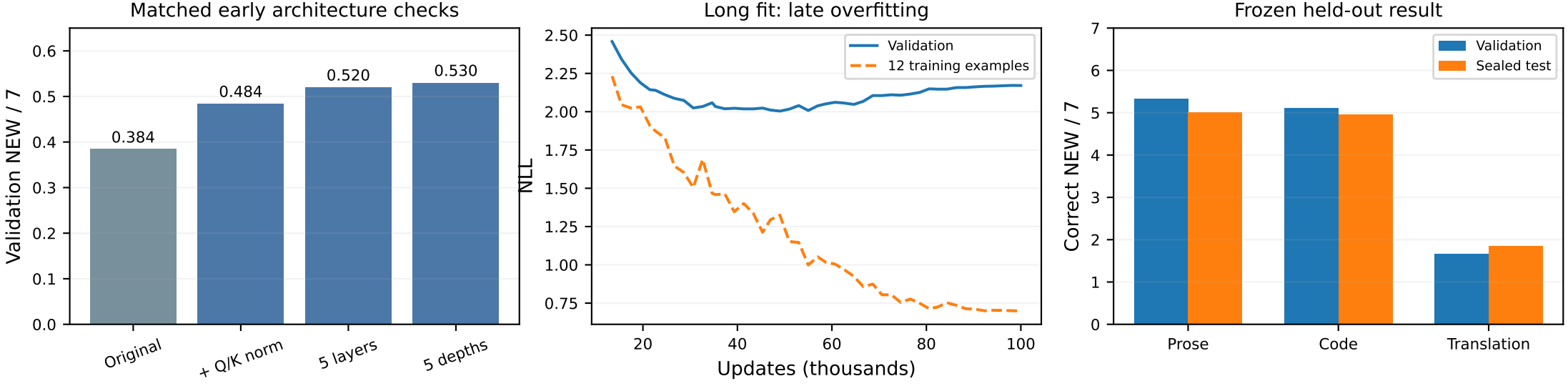
We trained a separate 79M-parameter, five-layer drafter from random weights for frozen Qwen2.5-0.5B. Following DFlash, it attends to five target hidden-state depths and predicts seven masked future tokens jointly after one known anchor. Only the drafter learns; target embeddings and vocabulary weights stay frozen.

What helped. Controlled early comparisons improved validation consecutive-token matches from 0.384 → 0.484 with query/key RMSNorm, then → 0.520 with five draft layers. Adding five feature depths reached 0.530, but that incremental benefit was uncertain. Correct target-generated labels, causal context masks, normalized conditioning, and verified generation made the pipeline credible; these implementation checks were not independent improvement ablations.

Data and response coverage mattered most. A matched 6K → 24K response study improved acceptance 0.57 → 1.02; a later 24K → 49K study reached 1.46. Longer teacher responses gave the largest subsequent gain: at 6K updates on the same 49K examples, 64 → 256-token caps improved long-context validation 1.76 → 2.96; short-context gains were smaller. The final large fits started fresh on 131K prose/code examples, then added translation to reach 196K, preserving optimizer state. Translation improved rapidly after inclusion. Multiple prose prefixes increased contexts, not independent articles.

What did not establish a win. Tiny datasets memorized almost perfectly but generalized poorly. More repetitions eventually worsened validation loss; select the earlier checkpoint. The 2B-target experiment underfit and did not improve drafting, with family/tokenizer/capacity confounds. Historical four-token experiments found no benefit from layer dropout/null memory; parallel prediction matched or beat diffusion. Probe-selected layers did not beat all-layer conditioning. Compressed top-25 Jacobian-lens conditioning worsened loss; it was not the paper's sparse J-space projection, and does not settle the new question. None of these older controls was repeated on the qualified drafter.

Outcome. The selected 70.6K-update checkpoint scores 4.03/7 on validation and 3.94/7 on the once-opened test: prose 5.01, code 4.96, translation 1.84; 189 usable prompts. All 24 longer rollouts matched Qwen exactly. Actual accepted draft tokens per verification were 4.51/4.30/1.20 by domain. These are one-seed target-imitation results; repetitive teacher outputs and changing data/exposure limit causal claims. Human answer quality and speedup were not established.



One seed; different experiments have different exposure budgets. Training diagnostic is tiny. Matching Qwen is not a human-quality score.

Sources: DFlash arXiv:2602.06036; Anthropic workspace/Jacobian lens; full reports and protocol in the repository.